Chapter 3: Conditional Statements

Python Programming Notes | Beginner Study Chapter

Making decisions

Conditional statements choose which code to run based on a condition. The if statement runs its block when its condition is True. elif checks another condition if earlier conditions were False. else runs when all earlier conditions were False. Only the first matching branch of an if/elif/else chain runs.

Comparison operators

Use == to compare equality, != for inequality, > for greater than, < for less than, >= for greater than or equal to and <= for less than or equal to. The assignment operator = assigns a value; it does not compare two values.

Logical operators

and is True when both conditions are True. or is True when at least one condition is True. not reverses a Boolean result. Parentheses can make combined conditions easier to read.

Example

A grading program can check whether marks are at least 80, then at least 50, and otherwise report that the student needs practice. Put a colon after each condition and indent the statements in its block.

Try this code

```
marks = 72
if marks >= 80:
    print("Excellent")
elif marks >= 50:
    print("Pass")
else:
    print("Needs practice")
```